

Subject: Cloud Computing

Practical no: 8

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Roll no: 15

Teacher Sign:

Q]. Containerize a simple web application using Docker and run it as a cloud service.

Aim: To containerize and deploy a simple Python web application using Docker.

Soln:

--app.py

```
from http.server import HTTPServer, BaseHTTPRequestHandler
```

```
class SimpleCloudApp(BaseHTTPRequestHandler):
```

```
    def do_GET(self):
```

```
        self.send_response(200)
```

```
        self.send_header(
```

```
            'Content-type',
```

```
            'text/html'
```

```
        )
```

```
        self.end_headers()
```

```
        self.wfile.write(
```

```
            b"<h1>Cloud Computing Practical: "
```

```
            b"Docker Service Running Successfully!</h1>"
```

```
        )
```

```
if __name__ == "__main__":
```

```
    server = HTTPServer(
```

```
        ('0.0.0.0', 8080),
```

```
        SimpleCloudApp
```

```
    )
```

```
    print(
```

```
        "[*] Containerized service "
```

```
        "listening on port 8080..."
```

```
    )
```

```
    server.serve_forever()
```

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--Dockerfile

FROM python:3.9-slim

WORKDIR /app

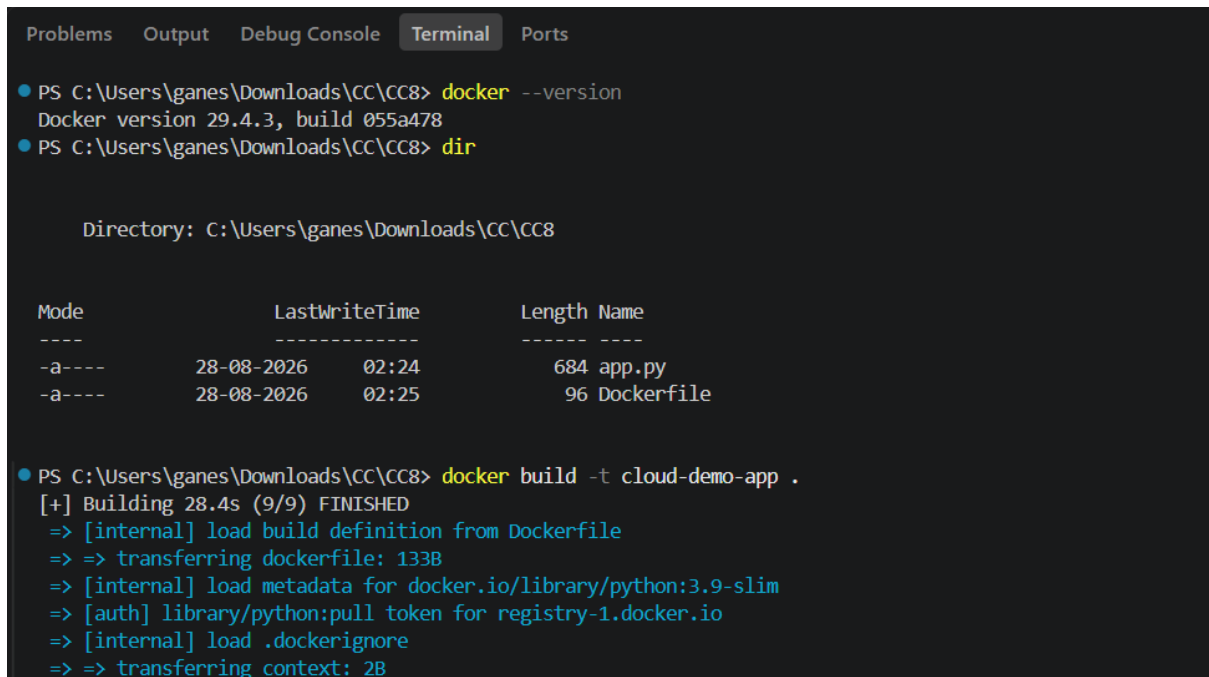
COPY app.py .

EXPOSE 8080

CMD ["python", "app.py"]

Terminal Commands :

OUTPUT :



```
Problems Output Debug Console Terminal Ports
● PS C:\Users\ganes\Downloads\CC\CC8> docker --version
Docker version 29.4.3, build 055a478
● PS C:\Users\ganes\Downloads\CC\CC8> dir

Directory: C:\Users\ganes\Downloads\CC\CC8

Mode                LastWriteTime         Length Name
----                -
-a----            28-08-2026    02:24           684 app.py
-a----            28-08-2026    02:25           96 Dockerfile

● PS C:\Users\ganes\Downloads\CC\CC8> docker build -t cloud-demo-app .
[+] Building 28.4s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 133B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
```

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```
PS C:\Users\ganes\Downloads\CC\CC8> docker run -d -p 8080:8080 --name cloud_container cloud-demo-app
411c16ea7cdb3b0b4e40418234dee3ff0f02a047e76a911748027f5719ea2c5f

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint cloud_
container (5b9a2efe9fe34f98b6ac0b858b096138604e41f5c0e050e2459de3edd83b5259): Bind for 0.0.0.0:8080 failed: port is already allocated

Run 'docker run --help' for more information
PS C:\Users\ganes\Downloads\CC\CC8> netstat -ano | findstr :8080
TCP      0.0.0.0:8080          0.0.0.0:0             LISTENING             11552
TCP      [::]:8080            [::]:0                LISTENING             11552
TCP      [::1]:8080           [::1]:0               LISTENING             15228
PS C:\Users\ganes\Downloads\CC\CC8> ^C
PS C:\Users\ganes\Downloads\CC\CC8> tasklist /FI "PID eq 11552"

Image Name                        PID Session Name        Session#    Mem Usage
=====
com.docker.backend.exe           11552 Console              2           2,04,460 K
PS C:\Users\ganes\Downloads\CC\CC8> tasklist /FI "PID eq 15228"

Image Name                        PID Session Name        Session#    Mem Usage
=====
wsirelay.exe                    15228 Console              2           10,424 K
PS C:\Users\ganes\Downloads\CC\CC8>
PS C:\Users\ganes\Downloads\CC\CC8> docker ps

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
62690c5aa832   adminer    "entrypoint.sh docke..." 4 weeks ago   Up 4 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
3a55b4e2258b   redis:7-alpine    "docker-entrypoint.s..." 4 weeks ago   Up 4 minutes  0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
8b74dafbe1e4   minio/minio    "/usr/bin/docker-ent..." 4 weeks ago   Up 4 minutes  0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001/tcp
1->9000-9001/tcp polyhub_minio
890c1c9b9f00   postgres/postgis:16-3.4    "docker-entrypoint.s..." 4 weeks ago   Up 4 minutes  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
polyhub_postgres

PS C:\Users\ganes\Downloads\CC\CC8> docker ps -a

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
411c16ea7cdb   cloud-demo-app    "python app.py"          2 minutes ago   Created
cloud_container
62690c5aa832   adminer    "entrypoint.sh docke..." 4 weeks ago   Up 4 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
polyhub_adminer
3a55b4e2258b   redis:7-alpine    "docker-entrypoint.s..." 4 weeks ago   Up 4 minutes  0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
polyhub_redis
8b74dafbe1e4   minio/minio    "/usr/bin/docker-ent..." 4 weeks ago   Up 4 minutes  0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001/tcp
polyhub_minio
890c1c9b9f00   postgres/postgis:16-3.4    "docker-entrypoint.s..." 4 weeks ago   Up 4 minutes  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
polyhub_postgres

PS C:\Users\ganes\Downloads\CC\CC8> docker rm cloud_container
cloud_container
PS C:\Users\ganes\Downloads\CC\CC8> docker run -d -p 8081:8080 --name cloud_container cloud-demo-app
4e760e57b21526f368b34dba2bb59ad13fba43501add039ce28e55f662216ce
PS C:\Users\ganes\Downloads\CC\CC8> docker ps

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
4e760e57b215   cloud-demo-app    "python app.py"          12 seconds ago   Up 11 seconds  0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp
cloud_container
62690c5aa832   adminer    "entrypoint.sh docke..." 4 weeks ago   Up 5 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
polyhub_adminer
3a55b4e2258b   redis:7-alpine    "docker-entrypoint.s..." 4 weeks ago   Up 5 minutes  0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
polyhub_redis
8b74dafbe1e4   minio/minio    "/usr/bin/docker-ent..." 4 weeks ago   Up 5 minutes  0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001/tcp
polyhub_minio
890c1c9b9f00   postgres/postgis:16-3.4    "docker-entrypoint.s..." 4 weeks ago   Up 5 minutes  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
polyhub_postgres
PS C:\Users\ganes\Downloads\CC\CC8>
```

localhost:8081

localhost:8081

Cloud Computing Practical: Docker Service Running Successfully!